Akoizo Original Practice - Form 3 - Answer key

Division C | Hovercraft | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Thrust is 0.30 N, friction and drag total 0.10 N, and mass is 1.0 kg. What is the acceleration?

A. 0.20 m/s^2

B. 0.30 m/s^2

C. 0.40 m/s^2

D. 0.10 m/s^2

Correct: A

2. (2 points) Which load gives the maximum Mass Score of 20?

A. All 4 rolls, the half roll, and 20 loose nickels

B. 4 full rolls only

C. 20 loose nickels only

D. The half roll and 20 loose nickels

Correct: A

3. (2 points) Where must the timing dowel be mounted?

Unit: Hitting the Target Time: calibration and repeatability.
Worked-case data: Your curve shows setting 3 -> 9.6 s and setting 4 -> 11.2 s (full load, 5.0 V). The Target Time is 10.0 s. Which setting do you choose?
Separate investigation observations:
Calibration: Settings 2, 4, 6 gave 8.0, 12.0, 16.0 s.
Target Time: 13.0 s announced.
Run 1 at 4.5: 13.5 s.

A. At the rear, at least 10 cm tall

B. On the thrust fan guard

C. Within 3.0 cm of the front edge, reaching at least 20 cm above the track

D. Anywhere on the vehicle

Correct: C

4. (2 points) Which component could legally slow a motor?

Unit: The electrical system: batteries, motors, and speed control.
Worked-case data: A team wants to use eight 1.5 V alkaline AA cells. Is that legal, and what are the options?
Separate investigation observations:
Monday: Fresh pack 5.2 V, knob at 3: 11.8 s.
Wednesday: Same knob, pack at 4.7 V: 13.4 s.
Wednesday, recharged: Pack 5.2 V, knob at 3: 11.9 s.

A. A laser timing gate

B. A lithium regulator chip

C. A potentiometer in series with the motor

D. An Arduino speed controller

Correct: C

5. (4 points) Explain in your own words why a hovercraft floats.

Unit: Levitation: the air cushion and the skirt.
Worked-case data: A hovercraft with nickels weighs 1.6 kg and has a 0.35 m x 0.35 m cushion. What cushion pressure is needed?
Separate investigation observations:
Empty test: Vehicle glides freely across the floor with lift on.
With all nickels: Rear skirt drags and the craft stops quickly.
Nickels moved forward 6 cm: Glides freely again, sitting level.

answer: A fan pushes air under the hull faster than it escapes, raising the pressure; that small extra pressure acting over the whole cushion area pushes up with a force equal to the weight. (4 points).

6. (4 points) List three things to bring to impound.

Unit: Impound, inspection, and competition day.
Worked-case data: A team’s vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?
Separate investigation observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

answer: The vehicle, spare parts, and calibration papers, with batteries stored separately. (4 points).

7. (4 points) Why calibrate both a full-load and a no-load setting?

Unit: Carrying nickels: mass, balance, and security.
Worked-case data: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.
Separate investigation observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

answer: Run time changes with mass, so having both lets the team switch loads between runs and still use a known, accurate setting. (4 points).

8. (4 points) Describe one way to keep a hovercraft running straight.

Unit: Thrust, friction, and motion down the track.
Worked-case data: A hovercraft must cross 185 cm in 12.0 s at steady speed. It enters the timed zone already at speed. What speed is required, and how does adding 0.2 kg affect it?
Separate investigation observations:
Trial 1: Start 5 cm behind line: 14.1 s.
Trial 2: Start 40 cm behind line: 11.6 s.
Trial 3: Start 40 cm behind, plus 2 rolls: 13.8 s.

answer: Align the thrust line with the center of mass and balance the load left to right so there is no turning moment. (4 points).

9. (4 points) What happens if a team member touches the vehicle during a run?

Unit: Scoring a run: Distance, Time, and Mass.
Worked-case data: TT is 12.0 s. A vehicle carries 3 rolls and 10 loose nickels and finishes in 13.2 s with nothing falling off. Find the run score.
Separate investigation observations:
Run plan A: No nickels, typical error 0.3 s at TT = 12 s, always completes.
Run plan B: All nickels, typical error 0.6 s, completes 9 times in 10.
Run plan C: All nickels with ramp, error 0.2 s.

answer: The run counts as complete but DS, TS, and MS are all zero, so it scores nothing. (4 points).

10. (4 points) Why is a Target Time near 26 s risky for a slow vehicle?

answer: A very slow vehicle can stall for 3 s or creep past 2 x TT, making the run incomplete with no Time Score. (4 points).

11. (6 points) Solve this case and explain your reasoning: A team wants to use eight 1.5 V alkaline AA cells. Is that legal, and what are the options?

reasoning: Eight cells in series: 8 x 1.5 = 12.0 V, exactly at the limit.
Two parallel strings of four: 6.0 V with double capacity.
Lithium or lead cells would be illegal regardless of voltage. (3 points).

conclusion: Eight in series is legal at 12.0 V, but leaves no margin; many teams prefer lower voltage with a potentiometer for control. (3 points).

12. (2 points) What caused the drag with the full load?

Supplied synthetic observations:
Empty test: Vehicle glides freely across the floor with lift on.
With all nickels: Rear skirt drags and the craft stops quickly.
Nickels moved forward 6 cm: Glides freely again, sitting level.

A. The fan was too strong

B. The track was too wide

C. The load sat too far back, tilting the craft and compressing the rear skirt

Correct: C

13. (2 points) What would have happened without the fix?

Supplied synthetic observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

A. Nothing; shielding is optional

B. The vehicle would not run and would earn participation points only

C. A 0.9 multiplier would apply

Correct: B

14. (2 points) Define “Nickel roll” in the context of Hovercraft.

answer: 40 nickels in a bank wrapper, about 200 g; counts as R = 1. (2 points).

15. (2 points) Define “Net force” in the context of Hovercraft.

answer: Thrust minus drag and friction; sets acceleration. (2 points).

16. (2 points) Define “Time Score (TS)” in the context of Hovercraft.

answer: 40 x (1 - |t - TT|/TT) for complete runs; 0 if incomplete. (2 points).

17. (2 points) Define “Precision” in the context of Hovercraft.

answer: How close repeated results are to each other. (2 points).

18. (2 points) Define “Voltage sag” in the context of Hovercraft.

answer: Drop in battery voltage as it discharges or under load. (2 points).